

Software Engineering & Agile Methodology

Question Bank (1, 2, 3, 4 Marks)

PART A: 1 Mark Questions

- 1 Define Software.
- 2 What is Software Engineering?
- 3 Who is a Software Engineer?
- 4 List one characteristic of good software.
- 5 What is Software Crisis?
- 6 What does SDLC stand for?
- 7 What is a Software Process?
- 8 What is the main idea of the Waterfall Model?
- 9 What is the key focus of the Spiral Model?
- 10 Define Agile methodology.
- 11 What is Extreme Programming (XP)?
- 12 When was the Agile Manifesto introduced?
- 13 What is a Sprint in Scrum?
- 14 What is a self-organizing team?

PART B: 2 Marks Questions

- 1 Differentiate between Software and Hardware.
- 2 State any two objectives of Software Engineering.
- 3 Write any two characteristics of good software.
- 4 Mention two causes of Software Crisis.
- 5 List the phases of SDLC.
- 6 Differentiate between Software Process and SDLC.
- 7 Explain the Incremental Model briefly.
- 8 When is the Prototyping Model used?
- 9 Write any two features of the Agile Model.
- 10 What is the role of Product Owner in Scrum?

PART C: 3 Marks Questions

- 1 Explain the importance of Software Engineering.
- 2 Explain Software Crisis in detail.
- 3 Describe the phases of SDLC with a neat diagram.
- 4 Explain the Waterfall Model with advantages and disadvantages.
- 5 Explain the working of the Spiral Model.
- 6 What is Agile methodology? Explain its benefits.
- 7 Describe the main practices of Extreme Programming.

- 8 Explain the 12 principles of Agile software development.
- 9 Explain the Scrum framework.
- 10 Explain the concept of self-organizing teams in Agile.

PART D: 4 Marks Questions

- 1 Compare Waterfall, Incremental, Prototyping, Spiral, and Agile Models.
- 2 Explain SDLC in detail with all phases and diagram.
- 3 Explain Agile Manifesto and its four values in detail.
- 4 Explain Scrum including roles, events, and artifacts.
- 5 Explain Extreme Programming with advantages and disadvantages.